Final project - Deep Learning Course - 2025-Semester B

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1 Introduction

Pneumonia is a potentially life-threatening lung infection that remains a significant global health concern. Early detection and diagnosis of Pneumonia are critical for effective treatment but diagnosis involves chest X-ray imaging which is interpreted by radiologists, a process that can be time-consuming, and dependent on the availability of experts.

In this project, we aim to build and compare two deep learning models: a Convolutional Neural Network (CNN) and a Vision Transformer (ViT) in order to classify chest X-ray images as either normal or indicative of pneumonia.

Our main goal is to investigate how modern transformer-based architectures, originally designed for natural language processing and later adapted for computer vision tasks, perform in medical image classification when compared to conventional CNN-based architectures.

1.1 Related Work

Several studies have applied CNNs to detect pneumonia from chest X-rays with high accuracy. For instance, Rajpurkar et al. (2017) introduced CheXNet[1] , a 121-layer DenseNet model trained on the ChestX-ray14 dataset, achieving radiologist-level pneumonia detection accuracy. CNNs like ResNet and VGG have also demonstrated strong performance across various X-ray classification tasks.

Transformers, on the other hand, have gained traction in vision tasks following the introduction of the Vision Transformer (ViT) by Dosovitskiy et al. (2020)[2], which showed that, with sufficient data and compute, ViTs can outperform CNNs on large-scale image classification tasks. The ViT architecture segments an image into patches and treats them as a sequence of tokens, leveraging self-attention mechanisms for global context modeling.

[1] Rajpurkar et al. *CheXNet: Radiologist-Level Pneumonia Detection on Chest X-Rays with Deep Learning.* 2017. URL: `https://arxiv.org/pdf/1711.05225`.

[2] Dosovitskiy et al. *An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale.* 2021. URL: `https://arxiv.org/pdf/2010.11929`.

More recently, the paper "When Vision Transformers Outperform ResNets Without Pretraining or Strong Data Augmentations" (Chen et al., 2021)[3] explores how ViTs can surpass ResNet performance even under constrained training settings, suggesting that ViTs may also be effective in smaller, more domain-specific datasets like medical imaging.

2 Methodology

In order to find a better classifier for the task of chest X-ray images (Normal vs. Pneumonia), we designed an experiment framework incorporating both classical CNN architectures and modern transformer-based models. We trained and evaluated all models under two conditions: 'from scratch' and 'with pretrained weights'. We then explored the effect of Sharpness-Aware Minimization (SAM) on the ability of both methods to generalize their findings.

2.1 Dataset

As instructed, we used the Chest X-Ray Pneumonia dataset from Kaggle[4]

The dataset is organized into three folders: train, val, and test. Each containing two sub-folders (NORMAL and PNEUMONIA).
We preserved the test set as untouched and used it only for final evaluation.
For validation during training, we used the original val folder split and also tested re-partitioning a portion of the train folder to achieve a better classifier.

Our image preprocessing included: Converting grayscale X-rays to 3-channel images (which was required for ViT) as well as resizing, horizontal flipping, normalization, and augmentation (adding contrast).

2.2 Architectures evaluated

2.2.1 Custom CNN

We designed, trained and evaluated a custom CNN to be designated as our control model. The results from this model will serve as the baseline to which we will compare the following architectures.

2.2.2 ResNet-18

We selected ResNet-18 as our second architecture due to the comparison made in the quoted paper by Chen et al. (2022). We evaluated both training methods: From scratch (starting with random weights) and Transfer-learning from the ImageNet-1K weights.

[3] Chen et al. *When Vision Transformers Outperform ResNets without Pre-training or Strong Data Augmentations*. 2021. URL: `https://arxiv.org/pdf/2106.01548`.

[4] Kaggle. *Chest X-Ray Pneumonia dataset*. URL: `https://www.kaggle.com/datasets/paultimothymooney/chest-xray-pneumonia`.

2.2.3 Vision Transformer

Finally, we used the ViT-B-16 (Vision Transformer, Base size(12 Layers, 86M params), using 16x16 patch embeddings) model. Here too we evaluated both training methods: From scratch (starting with random weights) and Transfer-learning from the ImageNet-1K weights.

2.2.4 Sharpness-Aware Minimization (SAM)

SAM was incorporated into selected experiments to evaluate its impact on generalization. For each model, we ran parallel experiments with and without SAM.

2.3 Experiments summary

Model	Training Mode	SAM	Threshold
Custom CNN	From scratch	Yes	Tuned
Custom CNN	From scratch	No	Tuned
ResNet18	From scratch	Yes	Tuned
ResNet18	From scratch	No	Tuned
ResNet18	Transfer-learning	Yes	Tuned
ResNet18	Transfer-learning	No	Tuned
ViT-B/16	From scratch	Yes	Tuned
ViT-B/16	From scratch	No	Tuned
ViT-B/16	Transfer-learning	Yes	Tuned
ViT-B/16	Transfer-learning	No	Tuned
Custom CNN	From scratch	Yes	Fixed
Custom CNN	From scratch	No	Fixed
ResNet18	From scratch	Yes	Fixed
ResNet18	From scratch	No	Fixed
ResNet18	Transfer-learning	Yes	Fixed
ResNet18	Transfer-learning	No	Fixed
ViT-B/16	From scratch	Yes	Fixed
ViT-B/16	From scratch	No	Fixed
ViT-B/16	Transfer-learning	Yes	Fixed
ViT-B/16	Transfer-learning	No	Fixed

Table 1: Experiments Matrix

3 Experiment results

Each experiment detailed in this section was conducted using two thresholding approaches:

1. **Recall-tuned threshold:** Chosen per model to achieve at least 95% recall, which is critical in clinical settings where false negatives are unacceptable.

2. **Fixed threshold:** Standard classification decision point (0.5) to measure the model's ability to generalize.

3.1 Results of Recall-Tuned Threshold experiments

At a threshold selected to prioritize recall (>= 0.95), a requirement aligned with the medical context of pneumonia screening, as we sought to optimize recall and "catch" more true positives, even at the cost of some false positives.

3.1.1 From-Scratch Training without SAM

Model	Params (M)	Time (min)	Threshold	Acc	Prec	Recall	F1	AUROC
CustomCNN	0.48	147.4	0.176	**87.18%**	**83.99%**	**98.21%**	**90.54%**	**96.01%**
ResNet18	11.18	32.1	0.220	84.13%	80.63%	98.21%	88.55%	92.42%
ViT-B/16	21.67	81.0	0.092	78.37%	75.55%	96.67%	84.81%	90.88%

Table 2: Models trained from scratch without SAM

3.1.2 From-Scratch Training with SAM

Model	Params (M)	Time (min)	Threshold	Acc	Prec	Recall	F1	AUROC
ViT-B/16	21.67	72.8	0.135	**86.22%**	**83.04%**	97.95%	**89.88%**	**94.30%**
ResNet18	11.18	6.6	0.351	84.29%	80.67%	**98.46%**	88.68%	93.12%
CustomCNN	0.48	31.1	0.342	78.04%	74.85%	97.69%	84.76%	92.00%

Table 3: Models trained from scratch with SAM

3.1.3 Transfer Learning without SAM

Model	Params (M)	Time (min)	Threshold	Acc	Prec	Recall	F1	AUROC
ResNet18	11.18	21.5	0.170	**86.54%**	**82.97%**	98.72%	**90.16%**	**94.93%**
ViT-B/16	85.80	78.6	0.176	84.62%	80.75%	**98.97%**	88.94%	94.41%

Table 4: Pretrained models fine-tuned without SAM

3.1.4 Transfer Learning with SAM

Model	Params (M)	Time (min)	Threshold	Acc	Prec	Recall	F1	AUROC
ViT-B/16	85.80	117.9	0.146	**85.10%**	**82.35%**	96.92%	**89.05%**	**92.98%**
ResNet18	11.18	32.0	0.154	78.21%	75.10%	**97.44%**	84.82%	90.73%

Table 5: Pretrained models fine-tuned with SAM

3.1.5 Global Comparison for Recall-Tuned Threshold

Model	Params (M)	Time (min)	Acc	Prec	Recall	F1	AUROC
CustomCNN (scratch)	0.48	146.81	**0.9006**	**0.8727**	**0.9846**	**0.9253**	**0.9578**
CustomCNN (scratch+SAM)	0.48	93.35	0.7660	0.7374	0.9718	0.8385	0.9074
ResNet18 (scratch)	11.18	45.24	0.8189	0.7844	0.9795	0.8712	0.9416
ResNet18 (scratch+SAM)	11.18	35.42	0.8317	0.7975	**0.9795**	0.8792	0.9267
ResNet18 (pretrained)	11.18	23.75	0.7933	0.7605	0.9769	0.8552	0.9156
ResNet18 (pretrained+SAM)	11.18	30.80	0.7821	0.7510	0.9744	0.8482	0.9073
ViT-B/16 (scratch)	21.67	87.47	0.8189	0.7844	0.9795	0.8712	0.9210
ViT-B/16 (scratch+SAM)	21.67	165.55	0.8558	0.8261	0.9744	0.8941	0.9332
ViT-B/16 (pretrained)	85.80	77.48	**0.8622**	**0.8276**	**0.9846**	**0.8993**	**0.9449**
ViT-B/16 (pretrained+SAM)	85.80	291.08	0.8510	0.8235	0.9692	0.8905	0.9298

Table 6: Summary of all recall-tuned experiments

3.2 Results of Fixed Threshold experiments

3.2.1 From-Scratch Training without SAM

Model	Params (M)	Time (min)	Threshold	Acc	Prec	Recall	F1	AUROC
CustomCNN	0.48	147.4	0.500	**91.35%**	**91.79%**	94.62%	**93.18%**	**96.01%**
ResNet18	11.18	32.1	0.500	86.70%	85.45%	**94.87%**	89.91%	92.42%
ViT-B/16	21.67	81.0	0.500	84.62%	91.29%	83.33%	87.13%	90.88%

Table 7: From-Scratch Training without SAM

3.2.2 From-Scratch Training with SAM

Model	Params (M)	Time (min)	Threshold	Acc	Prec	Recall	F1	AUROC
ViT-B/16	21.67	72.8	0.500	**88.46%**	**90.36%**	91.28%	90.82%	**94.30%**
ResNet18	11.18	**6.6**	0.500	87.98%	86.37%	**95.90%**	**90.89%**	93.12%
CustomCNN	0.48	31.1	0.500	83.01%	81.56%	94.10%	87.38%	92.00%

Table 8: From-Scratch Training with SAM

3.2.3 Transfer Learning without SAM

Model	Params (M)	Time (min)	Threshold	Acc	Prec	Recall	F1	AUROC
ResNet18	11.18	21.5	0.500	88.62%	**90.18%**	91.79%	90.98%	**94.93%**
ViT-B/16	85.80	78.6	0.500	**88.78%**	88.28%	**94.62%**	**91.34%**	94.41%

Table 9: Transfer Learning without SAM

3.2.4 Transfer Learning with SAM

Model	Params (M)	Time (min)	Threshold	Acc	Prec	Recall	F1	AUROC
ViT-B/16	85.80	117.9	0.500	**86.70%**	**89.87%**	**88.72%**	**89.29%**	**92.98%**
ResNet18	11.18	32.0	0.500	84.29%	87.63%	87.18%	87.40%	90.73%

Table 10: Transfer Learning with SAM

3.3 Global Comparison for Fixed Threshold experiments

Model	**Setup**	**Params (M)**	**Time (min)**	**Threshold**	**Acc**	**Prec**	**Recall**	**F1**	**AUROC**
CustomCNN	Scratch	0.48	**147.4**	0.500	**91.35%**	**91.79%**	94.62%	**93.18%**	**96.01%**
ResNet18	Scratch	11.18	32.1	0.500	86.70%	85.45%	**94.87%**	89.91%	92.42%
ViT-B/16	Scratch	21.67	81.0	0.500	84.62%	91.29%	83.33%	87.13%	90.88%
ViT-B/16	Scratch+SAM	21.67	72.8	0.500	88.46%	90.36%	91.28%	90.82%	94.30%
ResNet18	Scratch+SAM	11.18	**6.6**	0.500	87.98%	86.37%	**95.90%**	**90.89%**	93.12%
CustomCNN	Scratch+SAM	0.48	31.1	0.500	83.01%	81.56%	94.10%	87.38%	92.00%
ResNet18	Pretrained	11.18	21.5	0.500	88.62%	**90.18%**	91.79%	90.98%	**94.93%**
ViT-B/16	Pretrained	85.80	78.6	0.500	**88.78%**	88.28%	**94.62%**	**91.34%**	94.41%
ViT-B/16	Pretrained+SAM	85.80	117.9	0.500	**86.70%**	**89.87%**	**88.72%**	**89.29%**	**92.98%**
ResNet18	Pretrained+SAM	11.18	32.0	0.500	84.29%	87.63%	87.18%	87.40%	90.73%

Table 11: Global Model Comparison across Training Strategies (Scratch vs. Pretrained, with/without SAM)

4 Discussion

4.1 The Role of CustomCNN in our experiments

The **Custom CNN** was designed to serve as a lightweight control model to establish a baseline for our experimentation. While it showed surprisingly strong performance in specific settings (particularly when trained from scratch without SAM and evaluated at a fixed threshold of 0.5) it is important to clarify that this should not be interpreted as it being the top performer across the board.

In the recall-tuned experiments, for example, the Custom CNN reached high recall values only by lowering its threshold dramatically (e.g., down to 0.176), which can potentially introduce sensitivity to noise, lighting variations, and positional distortions often seen in real-world medical images. This dependency on an extremely permissive threshold suggests that its performance is less robust and less clinically practical, especially when compared to pretrained models that achieved similar or better recall at more conservative thresholds.

4.2 Thresholding in Model Comparison

A major challenge encountered during evaluation was the use of different thresholding strategies: some results were optimized to achieve a target recall (≥ 0.95), while others were reported at the conventional fixed threshold of 0.5. This dual setup was necessary to simulate both clinical priorities (where recall is critical) and general model performance (where standard thresholding is used).

However, this also introduces a critical caveat: because thresholds varied across models and scenarios, direct comparisons between metrics (especially accuracy, precision, and F1) are inherently imprecise. For example, a model achieving high precision at a threshold of 0.5 may appear to underperform in the recall-tuned context, not because of a true performance gap but due to a shift in the decision boundary to favor recall. This makes full one-to-one comparisons between experiments impossible unless all models are re-evaluated under the same threshold.

While we attempted to report the most meaningful configuration for each model under realistic constraints, we recognize that this limits the comparability and makes aggregated "winner" declarations across all scenarios problematic.

4.3 Impact of SAM Optimization

The introduction of Sharpness-Aware Minimization (SAM) yielded notable benefits for certain models, but also some regressions:

- ViT-B/16 (from scratch) benefited significantly from SAM, suggesting that sharpness-aware optimization can stabilize training for large transformer models in data-scarce domains like medical imaging.

- In contrast, ResNet18 and CustomCNN showed minimal or negative effects from SAM, particularly when pretrained weights were used. This could imply that SAM requires careful hyperparameter tuning (e.g., ρ) to be effective on pretrained or smaller-scale CNNs.

4.4 Pretraining vs. Training from Scratch

Pretraining consistently led to improvements in both ViT and ResNet models, particularly in training efficiency and generalization when evaluated at a fixed threshold. For example, ViT-B/16 (pretrained, no SAM) achieved the highest F1-score (91.34%) and competitive AUROC (94.41%) at the standard 0.5 threshold, outperforming both its from-scratch variant and ResNet18. This demonstrates that pretrained transformer-based models can generalize well even without additional optimization techniques like SAM, especially when trained on medical tasks with limited data.

For ResNet18, however, gains from pretraining were more modest. In some cases, training from scratch yielded comparable or better results, suggesting that this architecture may not benefit as much from pretraining like ViT.

4.5 Limitations and Future Work

- The varying threshold strategy introduced evaluation inconsistency that makes full direct comparison difficult. Future work could include calibrating all models using common validation techniques (e.g., ROC or PR curve analysis) to ensure consistent comparison points.

- Dataset size was limited, which particularly affects the generalizability of high-capacity models like ViT. Applying domain-specific pretraining or data augmentation could further improve robustness.

- We did not explore ensemble methods or techniques such as attention map visualization (e.g., Grad-CAM[5]) to evaluate model interpretability, a key factor in clinical AI.

- SAM hyperparameters were fixed across experiments. Exploring a grid search over ρ or using adaptive techniques could yield better performance, particularly for CNNs.

4.6 Conclusion

In conclusion, **pretrained ViT-B/16 without SAM** was the most effective model in high-recall medical diagnosis scenarios, balancing sensitivity with precision and AUROC. While the **Custom CNN** occasionally appeared to outperform others in specific metrics, this was largely due to threshold tuning and should not be interpreted as a dominant architecture. Our findings underscore the need to evaluate models under consistent conditions, and to be cautious in interpreting performance without standardization across thresholding strategies. SAM showed promising results for ViT models and deserves further investigation, particularly in conjunction with optimization and tuning.

5 Code

All of our experiments can be found in the accompanying Github repository: https://github.com/fouada/Deep-Learning-Project/

References

Chen et al. *When Vision Transformers Outperform ResNets without Pre-training or Strong Data Augmentations*. 2021. URL: `https://arxiv.org/pdf/2106.01548`.

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[5] Ramprasaath R. Selvaraju. URL: `https://arxiv.org/pdf/1610.02391`.